

# Slopes and local change

Project: Image-pattern Network Investigation

## 1. Your mission

### Your learning target

I can calculate an average rate of change over a stated interval.

I can estimate a local slope with nearby loss values.

I can explain the limits of a local direction clue.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Loss functions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Forward passes by hand. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Bias terms and decision boundaries. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Calculate an average rate of change over a stated interval

Estimate a local slope with nearby loss values

Explain the limits of a local direction clue

### Retrieve what you already know

For one example with prediction 4 and target 2, what is the squared error?

#### My starting explanation

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## This week's end goal

By the end, I can explain slopes and local change using a traced example and a stated limitation.

## 2. Notice the evidence

Let a one-example model predict  $w$  and let its target be 2. Its squared loss is  $L(w) = (w - 2)$  squared. Here  $w$  is the parameter we vary, not a new data example.  $L$  is a nonnegative function of that parameter.

**$w$  · Prediction minus target ·  $L(w)$**

**$w$ : 0**

**Prediction minus target: -2**

**$L(w)$ : 4**

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**$w$ : 1**

**Prediction minus target: -1**

**$L(w)$ : 1**

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**$w$ : 2**

**Prediction minus target: 0**

**$L(w)$ : 0**

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**$w$ : 3**

**Prediction minus target: 1**

**$L(w)$ : 1**

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**$w$ : 4**

**Prediction minus target: 2**

**$L(w)$ : 4**

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From  $w = 0$  to  $w = 1$ , the loss changes by -3 while  $w$  changes by 1. The average slope is -3. From  $w = 1$  to  $w = 2$  it is -1. The curve does not have one fixed slope everywhere.

**List the given facts and one fact that is not supplied.**

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### 3. See how it works

Near  $w = 1$ , use two symmetric nearby values: 0.9 and 1.1. Their losses are 1.21 and 0.81. Change in loss divided by change in  $w$  gives  $(0.81 - 1.21) / (1.1 - 0.9) = -2$ . This symmetric estimate describes change near the centre  $w = 1$ .

```
def loss(w):  
    return (w - 2) ** 2  
  
def local_slope(w, step):  
    if step <= 0:  
        raise ValueError("Step must be positive")  
    return (loss(w + step) - loss(w - step)) / (2 * step)  
  
for w in [1, 2, 3]:  
    print(w, round(local_slope(w, 0.1), 6))  
  
# Slopes near these centres: -2, 0, 2
```

For this quadratic, the symmetric estimate equals the exact slope  $2(w - 2)$  in exact arithmetic. Numerical estimates for other functions are generally approximate. Smaller steps can reduce some approximation errors, but extremely small steps can suffer from computer rounding.

### Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

## 4. Compare the cases

### Compare a familiar case with a changed case

At centre  $w = 0$ , use step 0.5. Calculate  $L(-0.5)$ ,  $L(0.5)$  and the symmetric slope.

Keep the procedure fixed. Explain what changed in the inputs and result.

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### Change one thing in the pictured case

Change: Move from 0 all the way to 5.

Prediction: Loss becomes 9, exceeding the starting 4.

Reason: A negative local slope suggests increasing a locally, not that every positive jump decreases loss.

## 5. Words that unlock the idea

### Word · Meaning

**Word:** Slope

**Meaning:** Change in output divided by change in input over a stated interval

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**Word:** Local change

**Meaning:** Change considered near a particular input or parameter value

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**Word:** Step size

**Meaning:** The amount by which an input or parameter is changed

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**Word:** Stationary point

**Meaning:** A point where a differentiable function has zero local slope

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Use two terms to explain a step in this week's example.

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## 6. Follow a worked example

### Compare slope clues on the same loss curve

**Current w · Local slope · Nearby downhill direction**

**Current w:** 1

**Local slope:** -2

**Nearby downhill direction:** Increase w a little

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**Current w:** 2

**Local slope:** 0

**Nearby downhill direction:** No first order direction

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**Current w:** 3

**Local slope:** 2

**Nearby downhill direction:** Decrease w a little

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These are local clues for  $L(w) = (w - 2)^2$ . At  $w = 2$  the displayed function has its minimum; zero slope does not establish a minimum for every possible function.

**At  $w = 3$ , use 2.9 and 3.1. Calculate both losses and the symmetric slope. Which small change should lower loss here?**

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### Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time.

Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

## A pictorial worked case: Slopes and local change

Use a local slope to propose a small move rather than claiming every move helps.

### The facts we will use

Toy loss  $J(a)=(a-2)^2$ . At  $a = 0$ , use symmetric step 0.5.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/9

## Slopes and local change

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



### 1. Evaluate nearby points

$J(0.5)=2.25$  and  $J(-0.5)=6.25$ .



### 2. Estimate slope

$\text{Slope}=(2.25-6.25)/(0.5-(-0.5))=-4$ .



### 3. Check small move

Move to  $a = 0.25$ : loss 3.0625, below starting loss 4.

### CHANGE ONE THING

Move from 0 all the way to 5.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

## Step · What happens and why

**Step:** 1. Evaluate nearby points

**What happens and why:**  $J(0.5)=2.25$  and  $J(-0.5)=6.25$ .

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**Step:** 2. Estimate slope

**What happens and why:**  $\text{Slope}=(2.25-6.25)/(0.5-(-0.5))=-4$ .

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**Step:** 3. Check small move

**What happens and why:** Move to  $a = 0.25$ : loss 3.0625, below starting loss 4.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

## Why this result follows

A slope describes how the loss changes near the current parameter. The symmetric calculation compares equal small moves on either side. Its negative result suggests a small increase may help at zero. It does not promise every increase helps: jumping to five passes the optimum at two and produces a larger loss. Check the new loss after each proposed step.

## What this example does not establish

A finite difference approximates local change; numerical precision and nonsmooth points need care in more complex objectives.

Read the labelled picture: Which calculation disproves “any increase helps”? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Change one thing: Move from 0 all the way to 5. Predict the result before checking the explanation. What remains the same?

## 7. Find and repair a mistake

A negative slope at  $w = 1$  suggests that a sufficiently small increase reduces this smooth loss locally. Moving to 1.1 lowers loss from 1 to 0.81. A jump all the way to 4 raises loss to 4. The initial slope cannot justify a step of any size.

A zero slope is also limited evidence. The nonnegative function  $F(w) = (w^2 - 1)^2$  has zero local slope at  $w = 0$ , yet  $F(0) = 1$  and  $F(0.1) = 0.9801$ . Small moves on either side lower that function. A flat first-order clue can occur at a point that is not a minimum.

## Locate the earliest unsupported step

Starting at  $w = 3$ , compare loss after moving to 2.9 and after moving to -1. Both moves decrease  $w$ .

**State one repair and the checks you would rerun.**

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## 8. Practise together

For  $L(w) = (w - 2)^2$ , calculate the average slope from  $w = 0$  to  $w = 2$ . Compare it with the local slope at  $w = 2$ .

**At  $w = 3$ , use 2.9 and 3.1. Calculate both losses and the symmetric slope. Which small change should lower loss here?**

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**Explain your prediction to a partner before comparing results.**

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### Practise with less help: Check a symmetric estimate

Use workbook W2 for the connected example “Slopes and local change”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

**Plan · Do · Check**

**Plan:** What is given? Which rule or procedure applies?

**Do:** Record each intermediate step.

**Check:** Does the result follow? What remains unknown?

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Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

## Check your reasoning

Use the worked case for Slopes and local change. Proposed explanation: "Divide by the step only". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

## Explain the picture

Which calculation disproves "any increase helps"? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

### My explanation using a pictured step and the given facts

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This is supported practice. Keep the separately named independent case for an attempt before feedback.

## 9. Run the investigation

### Predict, investigate and preserve the evidence

At centre  $w = 0$ , use step 0.5. Calculate  $L(-0.5)$ ,  $L(0.5)$  and the symmetric slope.

```
def loss(w):  
    return (w - 2) ** 2  
  
def local_slope(w, step):  
    if step <= 0:  
        raise ValueError("Step must be positive")
```

```

    return (loss(w + step) - loss(w - step)) / (2 * step)

for w in [1, 2, 3]:
    print(w, round(local_slope(w, 0.1), 6))

# Slopes near these centres: -2, 0, 2

```

## Record · Your evidence

**Record:** Given input and fixed rule

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Prediction before checking

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Actual result, or “not run”

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** First mismatch and one explanation

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Boundary or missing input and expected response

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

## Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

# 10. Build your understanding

## Explain → apply → challenge

For  $L(w) = (w - 2)^2$ , calculate the average slope from  $w = 0$  to  $w = 2$ . Compare it with the local slope

at  $w = 2$ .

Starting at  $w = 3$ , compare loss after moving to 2.9 and after moving to -1. Both moves decrease  $w$ .

For  $F(w) = (w^2 - 1)^2$ , compare  $F(0)$  and  $F(0.1)$ . Does zero local slope at zero prove a minimum?

**Explain which evidence supports your conclusion and which claim remains unproved.**

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## 11. Connect your project

Add a labelled slopes and local change evidence sheet to Image-pattern Network Investigation. Show how you can calculate an average rate of change over a stated interval. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

### Project evidence · Record

**Project evidence:** Component and version

**Record:** \_\_\_\_\_

\_\_\_\_\_

**Project evidence:** Inputs permitted at prediction time

**Record:** \_\_\_\_\_

\_\_\_\_\_

**Project evidence:** Trace and comparison with earlier work

**Record:** \_\_\_\_\_

\_\_\_\_\_

**Project evidence:** Failure, limit and human review

**Record:** \_\_\_\_\_

\_\_\_\_\_

## Evidence for your project

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

## 12. Show what you know

Always record the function, centre, nearby values and denominator used in a slope estimate. An average slope across a wide interval is not automatically the local slope at either endpoint. Do not compare slope magnitudes without considering parameter and loss units.

**A learner sees a negative estimated slope and says any larger parameter value will improve the loss. Give a counterexample using this lesson.**

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Next week changes one parameter at a time in a two-parameter model and combines those local slopes into a gradient.

For  $F(w) = (w^2 - 1)^2$ , compare  $F(0)$  and  $F(0.1)$ . Does zero local slope at zero prove a minimum?

**My limited claim and a useful next check**

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## 13. Study a complete case

**Complete evidence record**

**Compare slope clues on the same loss curve**

**Current  $w$  · Local slope · Nearby downhill direction**

**Current w: 1**

**Local slope: -2**

**Nearby downhill direction:** Increase w a little

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**Current w: 2**

**Local slope: 0**

**Nearby downhill direction:** No first order direction

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**Current w: 3**

**Local slope: 2**

**Nearby downhill direction:** Decrease w a little

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These are local clues for  $L(w) = (w - 2)^2$ . At  $w = 2$  the displayed function has its minimum; zero slope does not establish a minimum for every possible function.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

#### **My complete trace of the worked case**

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## **14. Try a new case independently**

### **Independent case: Estimate a new curve independently**

Let  $J(a) = (a + 1)^2$ . At  $a = 1$  use step 0.25 to find the symmetric slope. Compare  $J(1)$  with  $J(0.9)$ , then explain why moving to -4 does not follow safely from the slope clue.

#### **My independent explanation, trace and limits**

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Attempt this without hints

Let  $J(a) = (a + 1)^2$ . At  $a = 1$  use step 0.25 to find the symmetric slope. Compare  $J(1)$  with  $J(0.9)$ , then explain why moving to -4 does not follow safely from the slope clue.

#### My independent procedure, result and limitation

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Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

### Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

## 15. Your lesson route

Start with this question: Use a local slope to propose a small move rather than claiming every move helps.

### By the end, I can · How I will show it

**By the end, I can:** Calculate an average rate of change over a stated interval

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Estimate a local slope with nearby loss values

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Explain the limits of a local direction clue

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

## 16. Finish and return

Close your notes. Explain this goal in your own words: Calculate an average rate of change over a stated interval

### My deciding step, evidence and one remaining question

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If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

## 17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Use a local slope to propose a small move rather than claiming every move helps.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Move to  $a = 0.25$ : loss 3.0625, below starting loss 4.

1. Check small move
2. Estimate slope
3. Evaluate nearby points

#### My selected conclusion, deciding fact and explanation

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Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Check small move Move to  $a = 0.25$ : loss 3.0625, below starting loss 4. A slope describes how the loss changes near the current parameter. The symmetric calculation compares equal small moves on either side. Its negative result suggests a small increase may help at zero. It does not promise every increase helps: jumping to five passes the optimum at two and produces a larger loss. Check the new loss after each proposed step.

### Practice 2: explain the deciding evidence

Changed-case practice: Move from 0 all the way to 5. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. Loss becomes 9, exceeding the starting 4.
3. This one example guarantees correct results for every future input.

#### My selected conclusion, deciding fact and explanation

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Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Loss becomes 9, exceeding the starting 4. A negative local slope suggests increasing a locally, not that every positive jump decreases loss.

## Repair a missing building block

Review Loss functions (ai-5/8). Goal: Calculate per-example squared error and mean squared error. Short practice: Why must the report name the chosen loss? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Forward passes by hand (ai-5/6). Goal: Record scores and activations in dependency order. Short practice: Why should feedback identify  $h$  before correcting only  $y$ ? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Bias terms and decision boundaries (ai-5/3). Goal: Calculate a weighted score with a bias term. Short practice: Is  $B$  positive under the original or changed model? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

## Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

## Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.